

BC 201 ADVANCES IN BIOCHEMISTRY (3-1-2)**OBJECTIVE**

To provide a comprehensive view of the metabolic network system.

COURSE CONTENT**UNIT I**

Biochemical reaction mechanism- A thermodynamic concept; Molecular Control of Metabolism(Carbohydrate, Proteins); Molecular Control of Metabolism(Lipids, Nucleic acid);

UNIT II

Commercially important Natural Product/metabolite Biosynthesis-I; Commercially important Natural Product/metabolite Biosynthesis-II; Methods for metabolic characterization, Cellular communication and transport;

UNIT III

Cell division, Different phases of Cell cycle; Checkpoints in cell cycle, Programmed Cell Death; Cancer – Preliminary idea of Oncogene, Proto-oncogene and Tumor suppressor gene; Protein dynamics, Protein-Protein, Protein-ligand and Protein-DNA interactions, Protein engineering and design;

UNIT VI

Biology of Informational Molecules, Receptors and Bio signaling_I; Biology of Informational Molecules, Receptors and Bio signaling_II; Regulation of gene expression.

TEXTBOOK AND REFERENCE BOOKS:

1. Lehninger Principles of Biochemistry, Macmillan
2. *Biochemistry*,. W. H. Freeman . Jeremy M Berg, John L Tymoczko, and Lubert Stryer.;
3. Biochemistry, Willey, Donald Voet, Judith G. Voet
4. Lippincott's Illustrated Reviews: Biochemistry(Lippincott's Illustrated Reviews series) by Pamela C. Champe, Richard A. Harvey, and Denise R. Ferrier
5. Textbook of Biochemistry with Clinical Correlations by Thomas M. Devlin.,
6. Harper's Illustrated Biochemistry, by Victor W. Rodwell , David A. Bender , Peter J. Kennelly
7. Schaum's Easy Outline Of Biochemistry by Philip W. Kuchel